

VERSUS Project Design Report

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The ER Diagram for the VERSUS project is shown below.

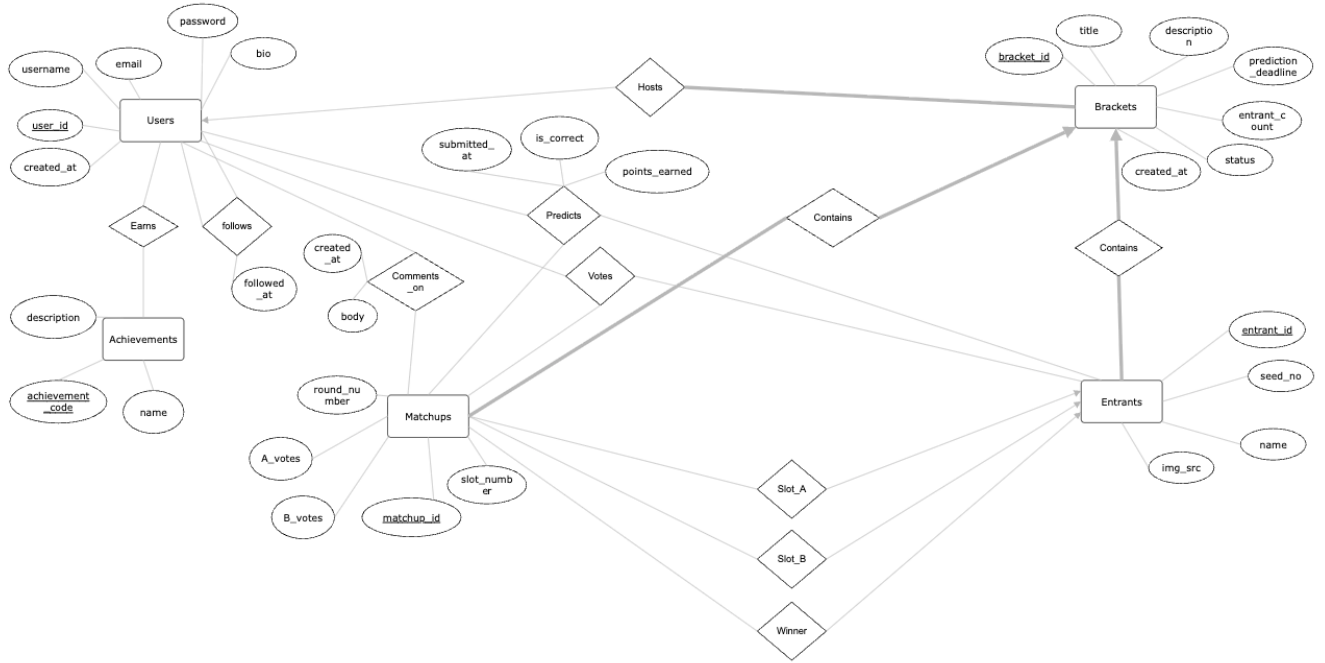


Figure 1: ER Diagram for the VERSUS Project

1 Entities

I opted to model only the primary project data, such as user information, bracket details as entities. Achievements are stored as a separate entity to allow for easy tracking and management of user accomplishments. The entities in the VERSUS project are as follows:

Entity	Attributes
User	user_id (PK), username, email, password, bio, created_at
Brackets	bracket_id (PK), title, description, prediction_deadline, entrant_count, status, created_at
Entrants	entrant_id (PK), seed_no, name, img_src
Matchups	matchup_id (PK), slot_number, round_number, A_votes, B_votes
Achievements	achievement_code (PK), name, description

2 Relationships

The rest of the project data and interactions between the entities are modeled as relationships because they represent the connections and dependencies between the entities in the system and

do not require their own entities (except for the ID attributes which I will add in the relational schema).

The relationships in the VERSUS project are as follows:

1. User **Hosts** Brackets.
 - Cardinality: 1:M (Full participation for Brackets) – a User can host multiple Brackets, but each Bracket must belong to a single User.
2. Brackets **Contains** Entrants
 - Cardinality: 1:M (Full participation for both Brackets and Entrants) – a Bracket must contain at least 4 Entrants and each Entrant must belong to a single Bracket.
3. Brackets **Contains** Matchups
 - Cardinality: 1:M (Full participation for both Brackets and Matchups) – a Bracket must contain at least 1 Matchup and each Matchup must belong to a single Bracket.
4. Entrant **Is In Slot A** in a Matchup
 - Cardinality: M:1 – an Entrant can be in slot A of 0 or many Matchups, but each Matchup can only have at most one Entrant in slot A (or NULL in the case of future Matchup).
5. Entrant **Is In Slot B** in a Matchup
 - Cardinality: M:1 – an Entrant can be in slot B of 0 or many Matchups, but each Matchup can only have at most one Entrant in slot B (or NULL in the case of future Matchup).
6. Entrant **Is A Winner Of** a Matchup
 - Cardinality: M:1 – an Entrant can be the winner of 0 or many Matchups, but each Matchup can only have at most one winner (or NULL in the case of future or unfinished Matchup).
7. User **Predicts** Entrant will win a Matchup
 - Cardinality: M:N:P – a User can predict the winner of 0 or many Matchups, and each Matchup can have 0 or many predictions from different Users. Similarly, an Entrant can be predicted as the winner of 0 or more matchups.
 - Attributes: `submitted_at`, `is_correct`, `points_earned`, `prediction_id` (I will add this in the schema).
 - Notes: I interpreted Predictions as a ternary M:N:P relationship instead of a separate entity because it keeps the ER diagram cleaner, avoiding the need for an additional entity and two binary relationships.
8. User **Votes For** Entrant in a Matchup
 - Cardinality: M:N:P – a User can vote for 0 or many Entrants in 0 or many Matchups, and each Matchup can have 0 or many votes from different Users. Similarly, an Entrant can be voted for in 0 or more matchups.

- Attributes: `vote_id` (I will add this in the schema).
- Notes: Similarly, I interpreted Votes as a ternary M:N:P relationship instead of a separate entity because it keeps the ER diagram cleaner.

9. User **Comments On** Matchups

- Cardinality: M:N – a User can comment on 0 or many Matchups, and each Matchup can have 0 or many comments from different Users.
- Attributes: `created_at`, `body`, `comment_id` (I will add this in the schema).

10. User **Earns** Achievements

- Cardinality: M:N – a User can earn 0 or many Achievements, and each Achievement can be earned by 0 or many Users.

11. User **Follows** Users

- Cardinality: M:N – a User can follow 0 or many other Users, and each User can be followed by 0 or many other Users.
- Attributes: `followed_at`
- Notes: this is a self-referencing relationship.

3 Relational Schema

The relational schema for the database is as follows:

- Users(`user_id`, `username`, `email`, `password`, `bio`, `created_at`)
- Brackets(`bracket_id`, `title`, `description`, `prediction_deadline`, `entrant_count`, `status`, `created_at`, `prediction_deadline`, `host_id` (FK))
- Entrants(`entrant_id`, `bracket_id` (FK), `seed`, `name`, `img_url`)
- Matchups(`matchup_id`, `bracket_id` (FK), `round`, `slot`, `votes_a`, `votes_b`, `entrant_a_id` (FK), `entrant_b_id` (FK), `winner_entrant_id` (FK))
- Predictions(`prediction_id`, `user_id` (FK), `matchup_id` (FK), `entrant_id` (FK), `submitted_at`, `is_correct`, `points_earned`)
- Votes(`vote_id`, `user_id` (FK), `matchup_id` (FK), `voted_for_entrant_id` (FK))
- Achievements(`achievement_code`, `name`, `description`)
- User_Achievements(`user_id`, `achievement_code`, `achieved_at`)

- Follows(follower_id, followed_id, followed_at)
- Comments(comment_id, user_id (FK), matchup_id (FK), created_at, body)

4 SQL Code

The SQL code for translating the relational schema into tables and constraints, and creating triggers is below:

DROP DATABASE IF EXISTS versus;

CREATE DATABASE versus;

USE versus;

```
CREATE TABLE Users (
    user_id      INT AUTO_INCREMENT PRIMARY KEY,
    username     VARCHAR(50)  NOT NULL UNIQUE,
    email        VARCHAR(255) NOT NULL UNIQUE,
    password     VARCHAR(255) NOT NULL,
    bio          TEXT,
    created_at   DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
);
```

```
CREATE TABLE Brackets (
    bracket_id   INT AUTO_INCREMENT PRIMARY KEY,
    host_id      INT NOT NULL,
    title        VARCHAR(255) NOT NULL,
    description   TEXT,
    entrant_count INT NOT NULL,
    status       ENUM(
        'draft',
        'predictions_open',
        'round_1', 'round_2', 'round_3', 'round_4', 'round_5',
        'completed'
    ) NOT NULL DEFAULT 'predictions_open',
    created_at   DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    prediction_deadline DATETIME,
    CONSTRAINT chk_entrant_count CHECK (entrant_count IN (4,8,16,32)),
    CONSTRAINT fk_brackets_host FOREIGN KEY (host_id) REFERENCES Users(user_id)
);
```

```
CREATE TABLE Entrants (
    entrant_id   INT AUTO_INCREMENT PRIMARY KEY,
    bracket_id   INT NOT NULL,
    seed         INT NOT NULL,
    name         VARCHAR(255) NOT NULL,
```

```

    img_url          VARCHAR(255),
    CONSTRAINT fk_entrants_bracket FOREIGN KEY (bracket_id) REFERENCES Brackets(bracket_id),
    CONSTRAINT uq_entrants_seed    UNIQUE (bracket_id, seed)
);

CREATE TABLE Matchups (
    matchup_id        INT AUTO_INCREMENT PRIMARY KEY,
    bracket_id        INT NOT NULL,
    round             INT NOT NULL,
    slot              INT NOT NULL,
    entrant_a_id      INT,
    entrant_b_id      INT,
    winner_entrant_id INT,
    votes_a           INT NOT NULL DEFAULT 0,
    votes_b           INT NOT NULL DEFAULT 0,
    CONSTRAINT fk_matchups_bracket FOREIGN KEY (bracket_id)
    REFERENCES Brackets(bracket_id),
    CONSTRAINT fk_matchups_a      FOREIGN KEY (entrant_a_id)
    REFERENCES Entrants(entrant_id),
    CONSTRAINT fk_matchups_b      FOREIGN KEY (entrant_b_id)
    REFERENCES Entrants(entrant_id),
    CONSTRAINT fk_matchups_winner FOREIGN KEY (winner_entrant_id)
    REFERENCES Entrants(entrant_id),
    CONSTRAINT uq_matchups_slot    UNIQUE (bracket_id, round, slot)
);

CREATE TABLE Predictions (
    prediction_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id      INT NOT NULL,
    matchup_id   INT NOT NULL,
    predicted_winner_id INT NOT NULL,
    submitted_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    is_correct   BOOLEAN,
    points_earned INT,
    CONSTRAINT fk_predictions_user FOREIGN KEY (user_id)
    REFERENCES Users(user_id),
    CONSTRAINT fk_predictions_matchup FOREIGN KEY (matchup_id)
    REFERENCES Matchups(matchup_id),
    CONSTRAINT fk_predictions_winner FOREIGN KEY (predicted_winner_id)
    REFERENCES Entrants(entrant_id),
    CONSTRAINT uq_predictions_user_matchup UNIQUE (user_id, matchup_id)
);

DELIMITER $$

```

```

CREATE TRIGGER prediction_open_check
BEFORE INSERT ON Predictions
FOR EACH ROW
BEGIN
    IF (
        SELECT status
        FROM Brackets
        WHERE bracket_id = (SELECT bracket_id FROM Matchups WHERE matchup_id = NEW.matchup_id)
        ) NOT IN ('predictions_open') THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Predictions are closed for this bracket';
    END IF;
END$$

```

```

CREATE TABLE Votes (
    vote_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    matchup_id INT NOT NULL,
    voted_for_entrant_id INT NOT NULL,
    CONSTRAINT fk_votes_user FOREIGN KEY (user_id)
    REFERENCES Users(user_id),
    CONSTRAINT fk_votes_matchup FOREIGN KEY (matchup_id)
    REFERENCES Matchups(matchup_id),
    CONSTRAINT fk_votes_entrant FOREIGN KEY (voted_for_entrant_id)
    REFERENCES Entrants(entrant_id),
    CONSTRAINT uq_votes_user_matchup UNIQUE (user_id, matchup_id)
);

```

```

CREATE TRIGGER vote_current_round_check
BEFORE INSERT ON Votes
FOR EACH ROW
BEGIN
    DECLARE current_round INT;
    SET current_round = (SELECT round FROM Matchups WHERE matchup_id = NEW.matchup_id);
    IF (
        SELECT status
        FROM Brackets
        WHERE bracket_id = (SELECT bracket_id FROM Matchups WHERE matchup_id = NEW.matchup_id)
        ) NOT IN (CONCAT('round_', current_round)) THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Voting is not open for this matchup';
    END IF;
END$$

```

```

CREATE TABLE Achievements (
    achievement_code ENUM('bracket_maker', 'locked_in') PRIMARY KEY,
    name VARCHAR(255) NOT NULL,

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```

        description TEXT NOT NULL
    );

CREATE TABLE User_Achievements (
    user_id INT NOT NULL,
    achievement_code ENUM('bracket_maker', 'locked_in') NOT NULL,
    achieved_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (user_id, achievement_code),
    CONSTRAINT fk_user_achievements_user FOREIGN KEY (user_id)
    REFERENCES Users(user_id),
    CONSTRAINT fk_user_achievements_achievement FOREIGN KEY (achievement_code)
    REFERENCES Achievements(achievement_code)
);

CREATE TRIGGER award_bracket_maker
AFTER INSERT ON Brackets
FOR EACH ROW
BEGIN
    DECLARE bracket_count INT;
    SELECT COUNT(*) INTO bracket_count FROM Brackets WHERE host_id = NEW.host_id;
    IF bracket_count = 10 THEN
        INSERT INTO User_Achievements (user_id, achievement_code)
        VALUES (NEW.host_id, 'locked_in');
    END IF;
END$$

CREATE TRIGGER award_locked_in
AFTER INSERT ON Predictions
FOR EACH ROW
BEGIN
    DECLARE prediction_count INT;
    SELECT COUNT(*) INTO prediction_count FROM Predictions WHERE user_id = NEW.user_id;
    IF prediction_count = 10 THEN
        INSERT INTO User_Achievements (user_id, achievement_code)
        VALUES (NEW.user_id, 'bracket_maker');
    END IF;
END$$

DELIMITER ;

CREATE TABLE Follows (
    follower_id INT NOT NULL,
    followed_id INT NOT NULL,
    followed_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (follower_id, followed_id),

```

```

        CONSTRAINT fk_follows_follower FOREIGN KEY (follower_id) REFERENCES Users(user_id),
        CONSTRAINT fk_follows_followed FOREIGN KEY (followed_id) REFERENCES Users(user_id),
        CONSTRAINT chk_no_self_follow CHECK (follower_id <> followed_id)
    );

CREATE TABLE Comments (
    comment_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    matchup_id INT NOT NULL,
    body TEXT NOT NULL,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT fk_comments_user FOREIGN KEY (user_id) REFERENCES Users(user_id),
    CONSTRAINT fk_comments_matchup FOREIGN KEY (matchup_id) REFERENCES Matchups(matchup_id)
);

```